



Accurate numerical simulation on the structural response of the VEGA payload fairing using modal coupling approach

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Abstract

Structural aeroacoustic interactions are of great interest in the design and manufacture of aerospace structures. During the lift off and the early phases of the launch various external loads like steady accelerations, random and broadband frequency vibrations, acoustic loads due to jet noise and fluid–structure interactions act on the lightweight panel structure of the payload fairing. Thereby internal vibrational loads and instability effects may be caused by the interaction of various mechanical loads and acoustic noise acting on the cylindrical shell structure. Within the project “Prediction of Acoustic Loads on Space Structures” funded by the European Space Agency (ESA) an aeroelastic coupling approach has been built up. Therein, the aerodynamic loads on the VEGA payload fairing (PLF) have been determined for different flight conditions by the use of the open-source CFD solver SU2. Due to the different discretization between the computational fluid dynamics (CFD) and computational structural dynamics (CSD) mesh the approach of Radial Basis Function (RBF) has been used to interpolate the resulting pressure distribution onto the structural model. According to that the dynamic response of the PLF has been analyzed by taking the basic aerodynamic forces, structural vibration and acoustic pressure fluctuations into account. An intrinsic part of this work is the numerical simulation and assessment of the interaction between structural vibrations, transonic flow and acoustic pressure fluctuations during the early launch phase.

Keywords Aeroelasticity · Structural response · VEGA payload fairing

1 Introduction

The payload fairing (PLF) is the upper conical part of a launch vehicle. The jettisonable structure protects the spacecraft or satellite against external impact like dynamic pressure, aerodynamic heating and acoustic interferences during atmospheric flight. Thereby internal vibration loads and instability effects may be caused due to the interaction of various mechanical loads and acoustic noise acting on the

shell structure. This can lead to enormous damage by transmission to the payload.

In addition, highly loaded PLFs are generally based on aluminum honeycomb cores with carbon fibre-reinforced plastic face sheets of various thickness and orientation as well as sound absorbing material for payload protection. The structural response is affected by several internal conditions like the preloaded state of structure, structural vibration and boundary conditions (BC) that have to be taken into account in the analysis and assessment of the structural response. Thus, there is a need for an efficient and accurate numerical procedure to solve the coupled aeroacoustic problem in a broad frequency range.

In the literature, some references related to the coupled structural response of spacecraft structures can be found. The aeroelastic analysis for the complete VEGA launch vehicle in transonic flight conditions has been performed by Capri et al. [5]. The technical book by Howe [6] gives basic physical principles of acoustic interaction. In the literature, several analytical and semi-empirical methods can be found to determine the turbulent boundary layer (TBL) wall

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pressure. Corcos [7] and Chase [8] proposed models for surface pressure spectra at low Mach number. Also Goody [9] developed an empirical model of the pressure fluctuations TBLs with good agreement to experimental data. A good literature survey about applied methods and experimental studies of pressure fluctuations has been given in [11]. Further analytical and computational aeroacoustic methods have been reviewed in [12] and [13].

In this paper, the influence of different loads acting on the elastic structure of the PLF of the European launch vehicle VEGA will be investigated. Therein the influence of different excitation forces as well as the preloaded state of the structure will be analyzed in terms of the structural response of the PLF.

On the basis of a weak aeroelastic coupling approach, the forced response of the linear PLF will be investigated by means of additional excitation forces. Initially, the steady aerodynamic forces acting on the PLF will be simulated to determine the preloaded state of the structure. The influence of the resultant change of the system matrices (e.g., mass and stiffness matrix) to the modal behavior and enforced motion of the PLF will be analyzed. Secondly, the structural response and resulting aerodynamic forces will be investigated with respect to additional pressure fluctuations acting on the linear elastic structure.

Therefore, an aeroelastic coupling process has been built up in MATLAB to analyze the coupled aero-structural behavior of the PLF. The process is based on a modal coupling approach. For the computation of the resultant deformations, the system matrices of the finite element model provided by the European Space Agency (ESA) have been generated by the use of NASTRAN DMAP alter. For the calculation of the aerodynamic forces, the open-source code SU2 [1] of Stanford University has been used.

2 Modeling approach

The used modeling approach for the structural acoustic response is based on a weak coupling for fluid–structure interaction. Thereby the influence on the flow field and the reverse influence on the structural deformation will be computed for each time step separately. For the analysis of structural deformations a NASTRAN finite element model provided by ESA has been considered. For the calculation of the aerodynamic forces the open-source code SU2 [1] of Stanford University has been used. Due to the different discretization of the structural and aerodynamic model the loads and deformations have been interpolated by the use of Radial Basis Functions (RBF) [2]. The use of the modal coupling approach permits an effective solution of the fluid–structure interaction problem. Different types of excitations can be added simply by MATLAB scripts and

there is close control of the specific physics. In addition, the modal coupling approach has good accuracy combined with high-computational speed. The complete simulation environment is implemented in MATLAB using modal coupling and will be described within this section.

2.1 Aeroelastic coupling

The interaction between the flow field and the elastic structure is based on a weak coupling for fluid–structure interaction. The aerodynamic forces have been calculated by the use of SU2 CFD code (see Sect. 2.3). For the prediction of the structural deformation of the PLF the system matrices of the finite element shell model generated by the use of NASTRAN DMAP alter have been used. The complete coupling procedure given in Fig. 1 is based on a modal coupling approach.

Therein the structural deformations are calculated separately by solving the ordinary differential equation (ODE) for each time step. For the interpolation of the resulting structural deformation, the approach of radial basis functions (RBF) has been used due to the different discretization of the CFD and structural mesh. The deformations of the aerodynamic mesh u_a are calculated from the structural deformations u_s by the use of the coupling matrix H , see [2].

$$u_a = H u_s. \quad (1)$$

These deformations are used as input for the SU2 CFD solver to compute the displacement-dependent aerodynamic forces within the coupling loop. Reciprocal the coupling matrix can be used to transform the aerodynamic forces f_a to the structural forces f_s using the transposed coupling matrix H^T [2].

$$f_s = H^T f_a. \quad (2)$$

Within this method the equilibrium of forces and momentum as well as the conservation equation of energy are conserved [3].

The general equation of motion in generalized coordinates for the analysis of the undamped dynamic response of the PLF is:

$$\{\ddot{q}(t)\} + [\Omega]\{q(t)\} = [\Phi]^T f_s \quad (3)$$

With respect to the RBF approach of Eq. (2), Eq. (3) can be rewritten as

$$\begin{aligned} \{\ddot{q}(t)\} + [\Omega]\{q(t)\} &= [\Phi]^T [H]^T f_a \\ &= [\Phi_a]^T f_a. \end{aligned} \quad (4)$$

The matrix $[\Omega] = [\omega_1^2, \dots, \omega_n^2]$ contains the eigenvalues of the system and the matrix $[\Phi]$ the structural mode shapes. The index ‘a’ denotes the structural mode shapes transformed to the aerodynamic surface mesh following Eq. (2).

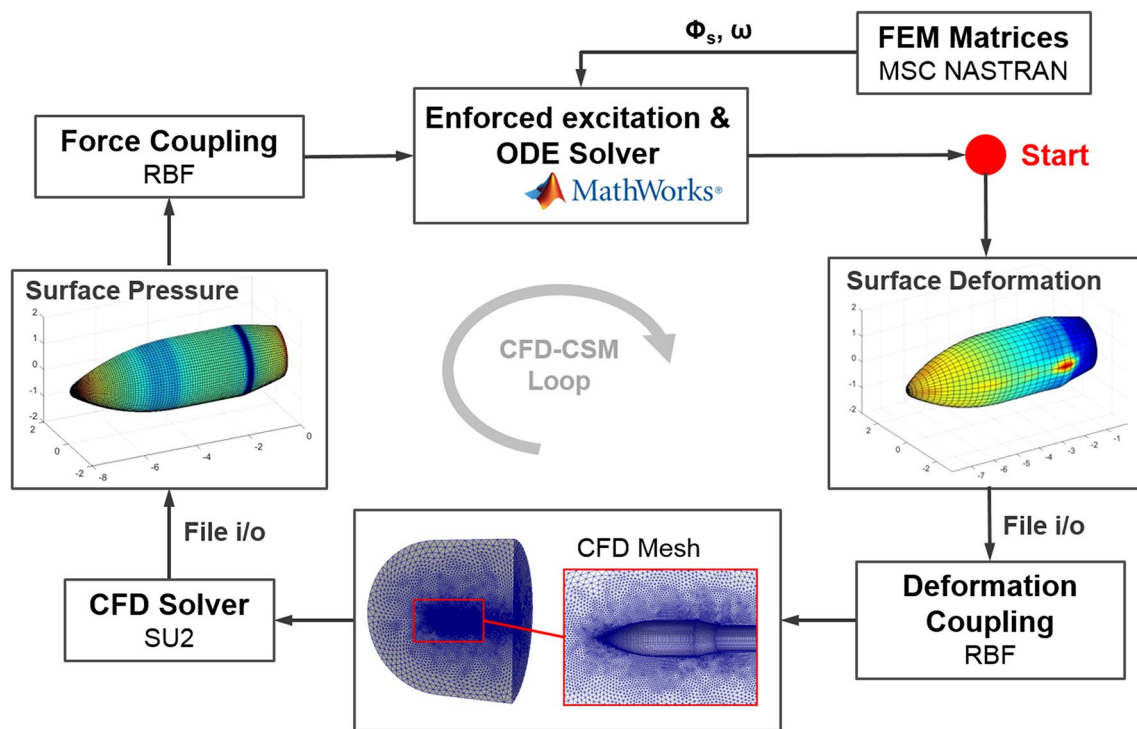


Fig. 1 Aeroelastic coupling procedure

The transformation has to be done only once, all continuing dynamic response analyses can be performed without further transformations [3].

In addition, the effect of damping has been neglected. Thus resonance effects are enhanced and might cause an overestimation of the vibrational amplitude. The structural response of different external and internal excitations will be used, and subsequently, the dynamic aeroelastic response of the structure can be performed to predict the combined influence of different load excitations. The numerical precision is mainly dependent on the step size of the coupling procedure and, respectively, the tolerance of the solution sequences.

2.2 Structural model

The structural model of the PLF has been provided by ESA in NASTRAN format and is modeled using anisotropic shell elements with stiffening beam elements within the separation plane. The bottom nodes are connected rigidly to the Attitude & Vernier Upper Module (AVUM), thus these nodes considered to be constrained by the RBE2 element in further analysis. For convenience only the payload fairing has been modeled. The lower part of the launch vehicle has been considered as rigid and is not part of the structural analysis.

In Fig. 2 a representative number of preloaded mode shapes of the finite element model is given. It can be clearly

seen that mainly the cylindrical part of the elastic shell structure deforms.

The structural vibrations induced by the jet engine of the launch vehicle during flight are applied to the AVUM by the use of transfer function. Thus the translational part of the rigid body motions can be taken into account. The change in angle of attack has been neglected due to infinite rigidity of the lower part of the launch vehicle.

The dynamic response computations are based on the system matrices (eigenvalues and eigenvectors) of the finite element model generated by the use of NASTRAN DMAP alter. To solve the ODE for each time step the Newmark method for linear systems has been used.

2.3 Aerodynamic model

The external aerodynamic forces induced by steady and unsteady flow conditions like regions of supersonic flow and shocks are simulated using the open-source code SU2 of Stanford University. The SU2 suite contains different solvers for the analysis of viscous and inviscid flow conditions using hybrid unstructured mesh topologies [1].

For the mesh generation, the Hypermesh software has been used. The geometric model contains the payload fairing and the 2nd and 3rd stage of the VEGA launch system to avoid numerical interaction on the elastic payload fairing. The CFD mesh, given in Fig. 3, is a hybrid

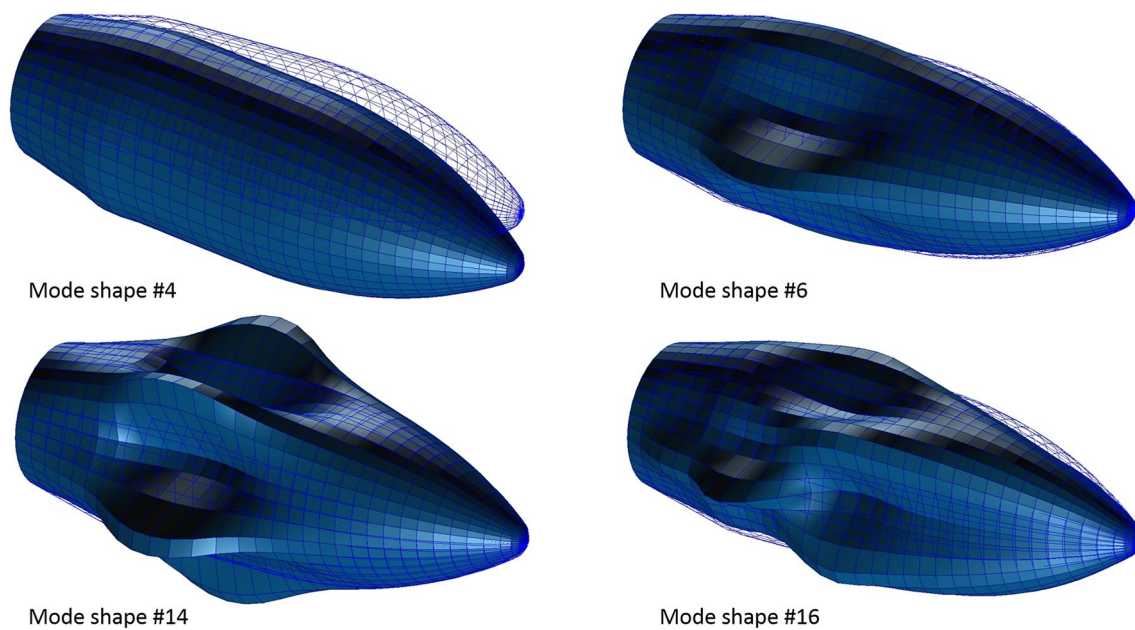


Fig. 2 Finite element model of the PLF and preloaded mode shapes 4, 6, 14 and 16

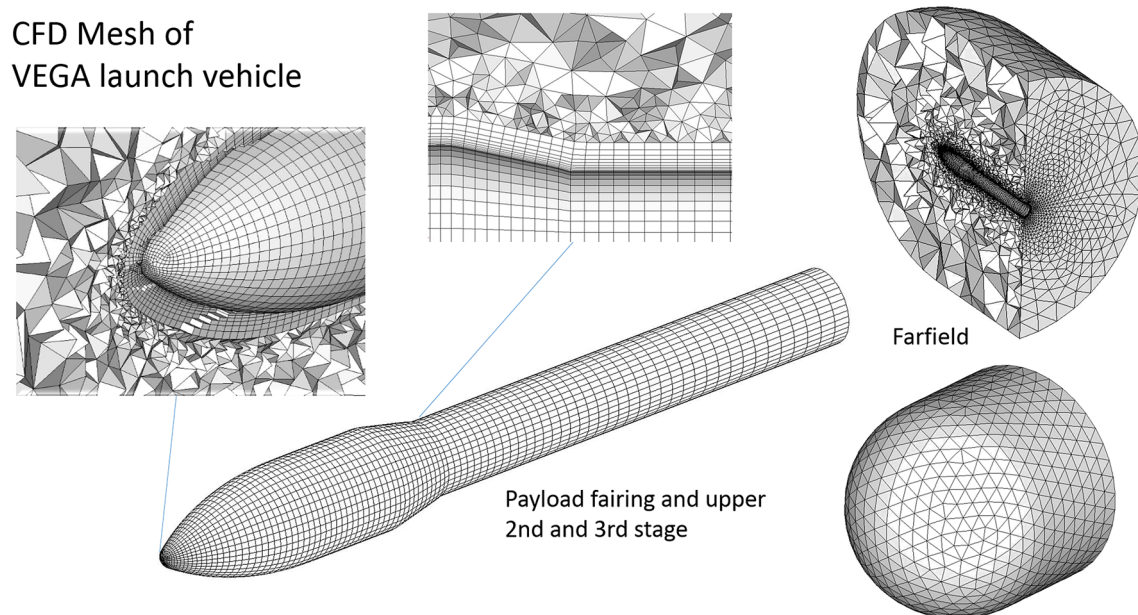


Fig. 3 Surface CFD mesh of VEGA payload fairing and 2nd and 3rd stage

unstructured mesh with 4000 structured surface elements on the surface walls and 220 thousand volume elements. For the preliminary calculations of the pressure distribution the inviscid Euler equations have been solved. This has been done to save computational time. In addition, due to neglecting the frictional forces the aerodynamic pressure is lower and hence the structural deformations

are stronger [10]. For the unsteady simulations, the dual time-stepping method with backward Euler scheme implemented in SU2 has been used. The positions of each node are updated following Eq. (1) and the differentiated grid values are computed using the finite differencing approximation of SU2 [4]. Subsequently, the specific changes of the deformed fluid volume mesh are computed for each

time step. The internal tolerance on the grid deformation has been set to 10^{-6} m.

3 Static aerodynamic results

The aerodynamic forces have been simulated by the use of the open-source code SU2. For the investigation of the aeroelastic behavior of the PLF different simulations of the model described above have been performed. Initially, the mean aerodynamic pressure on the PLF has been analyzed at an altitude of 6000 m and a freestream Mach number 0.72 (a) and 0.95 (b). The lower Mach case (a) has subcritical flow conditions whereas the upper Mach case (b) shows local shocks in the flow. The aim of these preliminary simulations was to ensure a stable numerical solution for the complex stream condition and a preliminary estimation of the aerodynamic loads acting on the elastic PLF.

3.1 Static pressure distribution

The steady flow distribution of the PLF given in Fig. 4 corresponds to the freestream Mach number $Ma = 0.72$ and a Reynolds number of $Re = 9.4 \times 10^6$. The streamwise pressure fluctuations, induced by compressibility effects are clearly formed. In the low transonic case, the Mach number is subcritical at the fairing cylinder. Downstream the fairing cylinder the velocity increase due to expansion-corner flow and the Mach number is close to unity. The flow recompresses directly after the kink [14, 15].

The upper Mach case (b) indicates a double shock system with subsonic flow between the first and second shock, given in Fig. 5. The selected flow corresponds to a Reynolds number of $Re = 12.4 \times 10^6$. The flow expands downstream the fairing cylinder and reaches supersonic flow conditions

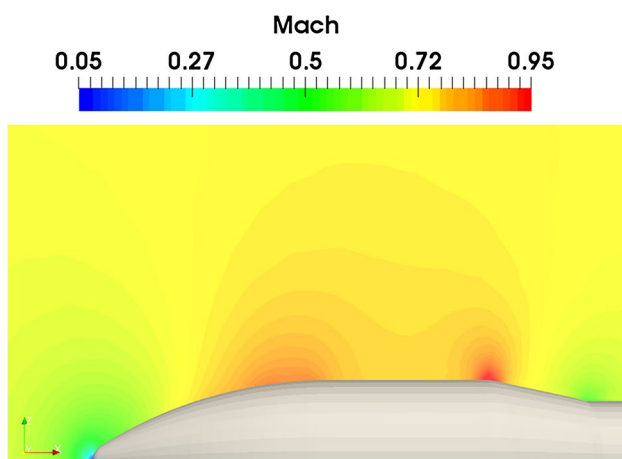


Fig. 4 Euler computation at Mach = 0.72

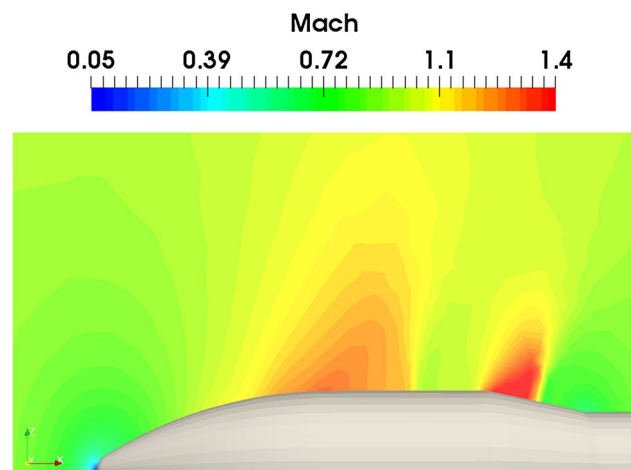


Fig. 5 Euler computation at Mach = 0.95

resulting in a compression shock. Further downstream the transonic expansion occurs upstream the kink and results to a second supersonic flow region around the corner with subsequent compression shock.

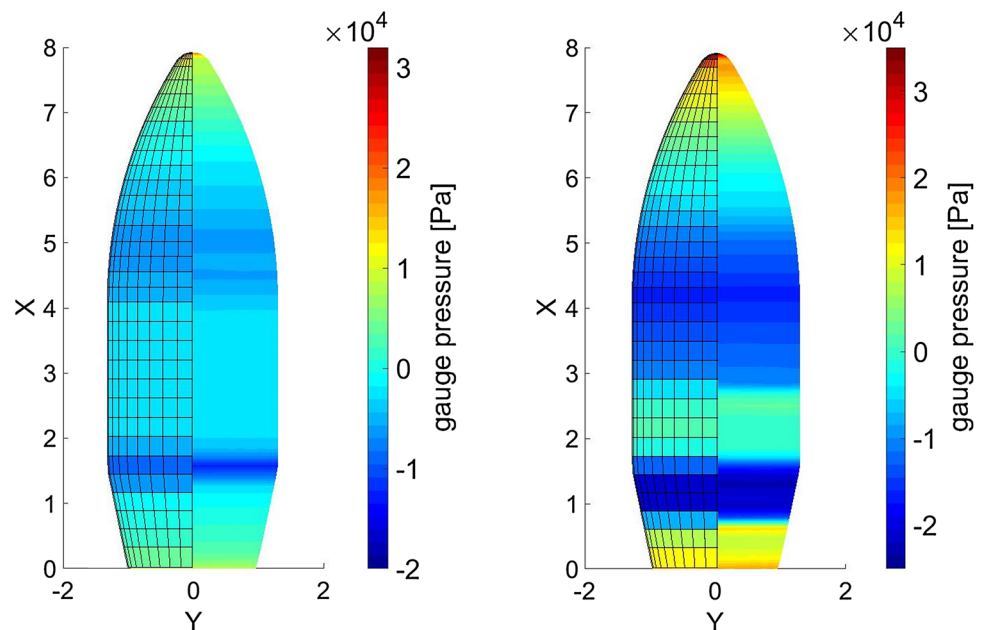
In Fig. 6 the resulting pressure distribution on the undeformed CFD mesh as well as on the structural mesh is given. In both Mach cases, the maximum pressure has been reached at the top of the PLF in the stagnation point. Within the lower Mach case $Ma = 0.72$ (cf. Fig. 6(left)) the first minimum pressure has been reached at the upper cylindrical part of the PLF due to the increasing stream velocity. Following that, the stream velocity decelerates downstream and the pressure gradient on the surface increases. The second pressure minimum of 20 kPa has been reached at the bottom kink of the PLF. There the critical Mach number has been reached.

The upper Mach case $Ma = 0.95$ (cf. Fig. 6(right)) indicates a double shock system with subsonic flow between the first and second shock. The resulting pressure distribution shows distinct regions of low-pressure and an increasing pressure gradient at the bottom of the PLF. The gauge pressure is in the order of magnitude of 25 kPa.

The transformation of the narrow pressure fluctuations in streamwise direction on the structural grid is of lower quality due to the rough resolution of the structural grid. In particular, the sharp-edged low-pressure region cannot be mapped to the structural grid. The computed pressure distribution on the CFD mesh with half the width of the structural panel size has been smoothed to the complete structural panel element through the pressure transformation (cf. Fig. 6). This behavior can also be seen at the curved section where narrow streamwise pressure fluctuations are located at the top of the PLF.

Due to the different discretization of the CFD and structural mesh the transformation of the aerodynamic pressure

Fig. 6 Pressure distribution at Mach = 0.72 (left) and Mach = 0.95 (right)



distribution has been performed by the use of Eq. (2). In the process the global equilibrium of forces is conserved. Never the less the sharp edged low-pressure region at the bottom PLF cannot be transferred to the structural mesh and is smeared on the structural mesh. The resulting error in the static pressure distribution has been neglected during further analysis.

4 Coupled simulation of the PLF

Initially the steady structural response of the PLF has been analyzed. Within the analysis the inner pressure of the PLF assumed to be the ambient pressure of the particular flight altitude. The static pressure distribution (cf. Sect. 3.1) has been added on the surface of the elastic structure by the use of RBFs. The resulting structural deformations have been transformed to the aerodynamic mesh (cf. Eq. 1) and subsequently, the aerodynamic pressure distribution of the deformed state has been computed. In addition, the dynamic response of the PLF has been simulated to validate the aeroelastic coupling procedure by comparison to NASTRAN.

4.1 Static deformation

The steady deformation of the elastic structure is given in Fig. 7. The PLF is compressed due to the dynamic pressure and the cylindrical part has been deformed outwards due to the distinct region of low-pressure. The deformation of the PLF is in the order of 1.0×10^{-3} m and is symmetric with respect to the separation plane, due to the corresponding beam elements.

In addition, the resulting pressure coefficient ΔC_p respectively on the undeformed and deformed structure is given in Fig. 8. Therein the values of the undeformed structure (bold dashed black line) have been compared to the values of the deformed structure (bold red line). The pressure coefficient denotes the ratio between the relative pressures $p - p_\infty$ of the flow field and the dynamic pressure $\frac{1}{2} \rho_\infty v_\infty^2$. The deformation of the PLF has been scaled for display reason. Within Fig. 8 the slice in the x - y -plane has been displayed. Therein only minor differences can be seen between the pressure distributions of the undeformed structure (black dashed line) in comparison to the pressure distribution of the deformed state (red line) of the PLF. Only small differences can be stated out in the subsonic flow region between the first and second shock.

The pressure disturbance at the bottom part of the launch vehicle is not part of the analysis because the structure has been assumed to be rigid. The steady solution of the deformed state has been used as initial input for the unsteady simulations.

4.2 Dynamic structural response

First, the coupling procedure has been used to analyse the structural response of the PLF by harmonic base excitation without external aerodynamic forces. In addition, the simulation has been performed by the use of the commercial software NASTRAN.

The case has been set to a harmonic excitation with 120 Hz. The frequency has been chosen to get a significant structural response, on the one hand and different modal participation beside the rigid body motion on the other.

Fig. 7 Static deformation at Mach = 0.72 (left) and Mach = 0.95 (right)

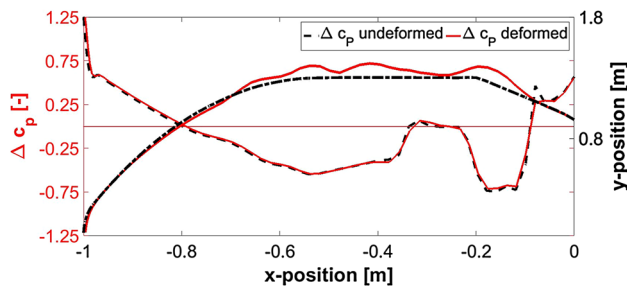
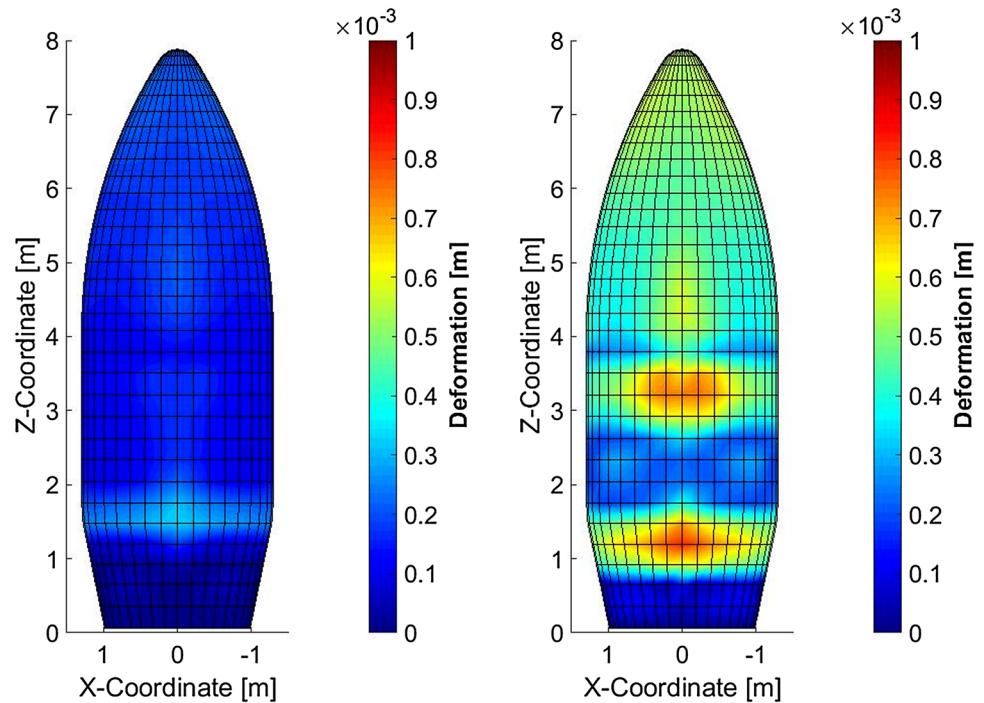


Fig. 8 Steady PLF deformation at $Ma = 0.95$

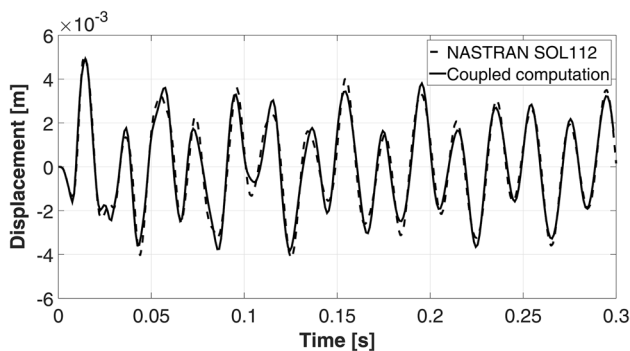


Fig. 9 Dynamic tip cone response under harmonic excitation

The excitation amplitude at the bottom point has been set to $\Delta y = 0.002$ m.

Figure 9 shows the displacement of the PLF tip cone as a function of time. The dynamic response by the use of the

coupling procedure (solid line) is in good agreement with the results obtained by the modal transient response analysis using NASTRAN (dashed line). Thus, the coupling procedure is able to predict the dynamic response of the PLF in the frequency range of interest.

5 Dynamic aeroelastic response

Furthermore, a coupled dynamic simulation has been performed as a plausibility check of the results. Therefore, a forced response analysis has been performed by global movement of the PLF along the AVUM attachment. Within the analysis of the structural response of the PLF the global movement of the launch vehicle, e.g., bending modes, has been neglected. The enforced motion in y-direction has been applied directly to the PLF at the bottom nodes connected rigidly to the AVUM. In addition, the PLF has no rotational degrees of freedom.

5.1 Harmonic excitation

The time history of the unsteady simulation has been performed within three intervals. The first time interval without excitation in order to ensure equilibrium conditions. Afterwards, a harmonic excitation with an amplitude acceleration of $2g$ started and subsequently stops after a period of eight cycles (cf. Fig. 10). The low sinusoidal frequency vibrations are in the range of 1–125 Hz [16]. Thus for the present simulation an excitation frequency of 50 Hz has been used.

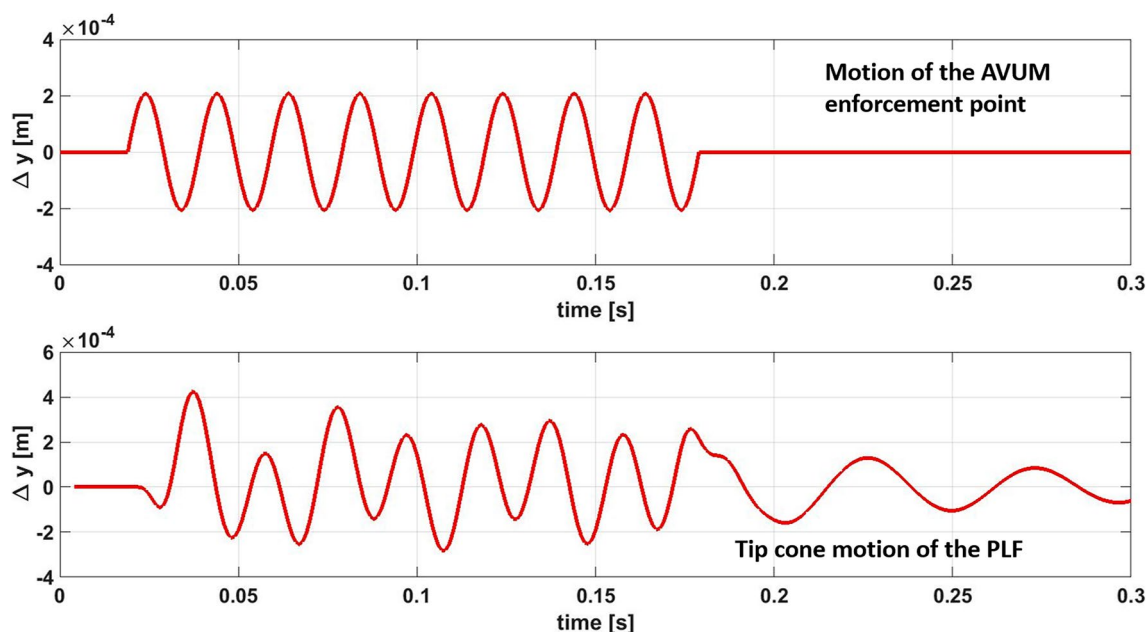


Fig. 10 Time history of enforcement point (upper) and response amplitude of the PLF tip cone (lower)

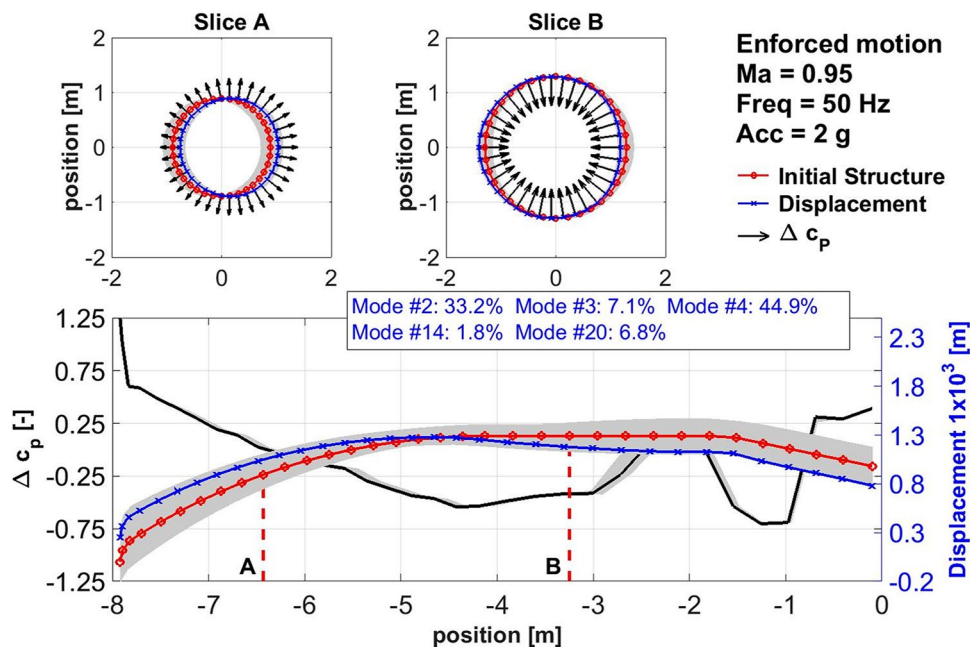
This artificial excitation leads still to a significant structural response, but avoids resonance effects due to neglect of damping. The response of the elastic PLF due to the acceleration has been determined with respect to the aerodynamic response.

The dynamic response of the elastic model is given in Fig. 11. There the dynamic response of the PLF has been simulated at the Mach case $Ma = 0.95$. The undeformed state (red line) of the PLF structure is shown as a meridian

slice at the bottom and as a cross-sectional view A and B at the top of Fig. 11. In addition, the scaled deformed state (blue line) of the structure is displayed. The grey background indicates the movement area of the elastic structure during the complete unsteady simulation.

One can see a slight ovalization of the cylindrical structure (cf. cross-sectional view A and B) caused by the steady pressure acting on the PLF and the stiffening beam elements within the separation plane. In addition, the enforcement of

Fig. 11 Simulation of the dynamic response using enforced motion at the PLF AVUM



the rigid body mode leads to the response of the structure. Thereby the major responding mode with a participation factor of 0.45 is the first bending mode. The subsequent mode with a participation factor of 0.33 is the enforced rigid body mode in y -direction. All additional participating mode shapes have a participation factor below 0.07.

Additionally, the pressure coefficient ΔC_p along the PLF is given in Fig. 11. Only minor differences can be seen on the pressure coefficient up to the first recompression shock, there a distinct shock motion appears while the second shock remains nearby unchanged. This is mainly due to the higher deformations at the particular elastic part of the PLF as well as the nearby undeformed area at the second recompression shock. All in all the pressure varies between 0.3 and 12 kPa.

6 Acoustic pressure

The acoustic load generation by supersonic jets and the impact on the structural components have been analyzed in detail by Koh et al. [18]. There the computational aeroacoustics (CAA) method has been used to determine the acoustic fluctuations.

6.1 Sound source near the jet nozzle

The mean pressure contours and the overall sound pressure level (OASPL) are presented in Fig. 12. The pressure contours shown in the $y > 0$ domain indicate the shock structure inside the supersonic jets. The freestream Mach number

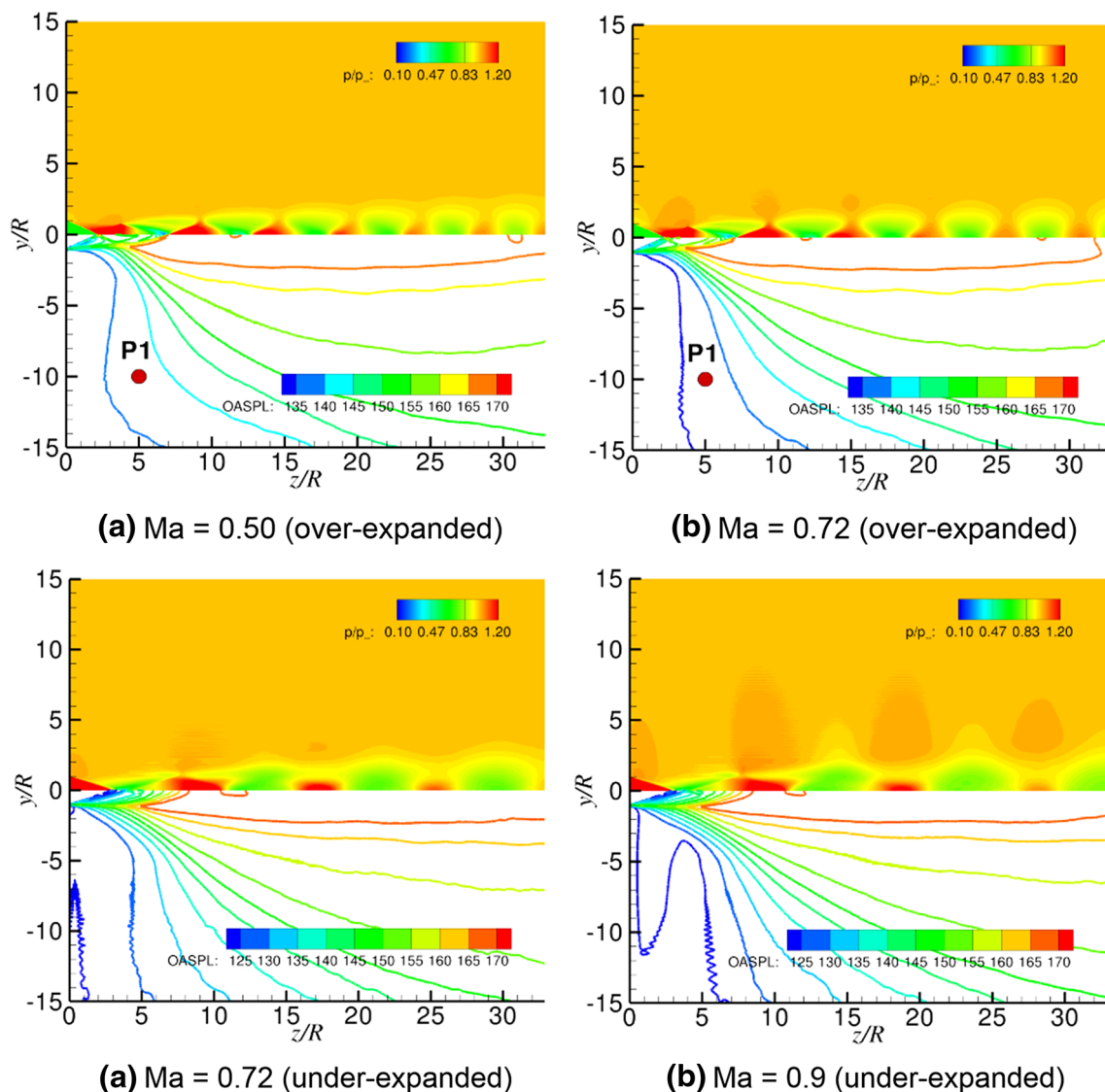


Fig. 12 Mean static pressure contours and overall sound pressure level at several freestream Mach numbers

possesses a serious impact on the pressure fluctuations in the jet near-field. At $y/R = 10$ and $z/R = 5$, i.e., $P1$ in Fig. 12a, b, the OASPL is 142.4 dB at the freestream Mach number 0.5 and 137.2 dB at the freestream Mach number 0.72. The same acoustic attenuation is reached by doubling the microphone distance from the acoustic source.

The wave propagation to the upstream direction excites the shear layer near the nozzle exit such that a feedback mechanism sustains the strong aerodynamic oscillation usually accompanied by its harmonics, usually described by the Strouhal number fD/U_j . The first harmonics of the supersonic jet have the frequency range $0.05 \leq fD/U_j \leq 0.2$ such that Tam et al. [17] found the fundamental screech tone valid for hot and cold jets (Eq. 5).

$$(fD)/U_j = \frac{0.67}{\sqrt{Ma_j^2 - 1}} \left[1 + \frac{0.7Ma_j}{\sqrt{1 + \frac{\gamma-1}{2}Ma_j^2}} \left(\frac{T_j}{T_{j0}} \right)^{-1/2} \right]^{-1}. \quad (5)$$

Here γ is the ratio of specific heat, Ma_j the Mach number of the jet, D_j and U_j respectively the diameter and velocity of the jet and the ratio T_j/T_{j0} of ambient to reservoir temperature of the jet. Eq. (5) is especially useful for the estimation of the screech tone of hot jets [17]. Based on this semi-empirical model the acoustic signal acting on the outer PLF has been determined up to the 5th harmonic frequency. The amplitude of the pressure fluctuations is based on the OASPL in Fig. 12. The first harmonic frequency for the hot jet has been located at a Strouhal number of $(fD/U_j) \approx 0.1$ corresponding to 100 Hz.

7 Summary and conclusion

Within this paper, the modal coupling approach has successfully been used to simulate the dynamic response of the elastic payload fairing. The resulting coupling procedure has been built up in MATLAB to analyze the coupled aerostuctural behavior of the PLF. The system matrices have been exported by the use of NASTRAN DMAP ALTER and have been implemented in the coupling environment. For the calculation of the aerodynamic forces, the open-source code SU2 of Stanford University has been used. Due to the different discretization of the structural and aerodynamic model the loads and deformations have been interpolated by the use of radial basis functions.

The static deformation of the elastic structure is symmetric with respect to the separation plane and has been simulated in different transonic flight regimes between Mach number $Ma = 0.72$ and $Ma = 0.95$. In the case of steady aerodynamic simulation, only small differences can be stated out in the transonic flow region and the

deformation due to the steady pressure distribution is small. In addition, the non-linear aerodynamic forces are significantly effected by the dynamic structural response, distinct shock motion and pressure variation can be seen due to the global movement of the PLF along the AVUM attachment. This might cause local deformations on the PLF during atmospheric flight and need to be analyzed in further detail.

Future work will concentrate on a detailed analysis of the dynamic response of the aeroelastic system in a coupled FSI simulation. Therein the acoustic pressure fluctuations will be added to the structural model by a broadband signal determined by the CAA superimposed by a full set of structural interface vibration. Therewith the combined influence of acoustic loads, unsteady aerodynamic loads and structural vibration can be determined in the aeroelastic coupling chain. There are currently no experiments foreseen to crosscheck the results of this theoretical aeroelastic computational work.

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